NT-3 Light: Behavior by Network Configuration

This document describes how the NT-3 light firmware behaves today, read from the source code. It covers not joined, Adaptiv and Daintree networks, and the difference between *joined* and *joined and grouped*.

- **Firmware:** branch `occupancy_improvement`, commit `8efba86` (includes the new Adaptiv group occupancy model, the Daintree level mapping and the LED/light-sensor changes).
- **Source of truth:** the code. Where this document says "today", that is what the code does, not necessarily what was intended. Section 11 lists behaviors that look unintended or worth a decision.
- **Levels** are Zigbee/PWM levels 0–255 unless a percentage is given (255 = 100 %, 204 = 80 %, 51 = 20 %, 25 = 10 %). The level control plugin's maximum level is configured as 255.

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1. Configurations

1.1 Network types

The light decides which network it is on by asking the coordinator for its IEEE address after joining, and checking the manufacturer prefix (OUI). It retries every 5 s until it knows. The result is saved in flash (`TOKEN_APP_WHICH_NETWORK`), so it is known at power up.

Network type	How it is detected	Code name
Not joined	Not on any network	`eNO_NWK`
Adaptiv	Coordinator IEEE starts with `8C:1F:64`	`eADAPTIV_NWK`
Daintree	Coordinator IEEE starts with `00:22:81`	`eDAINTREE_NWK`
Unknown	Joined, but the coordinator is neither	`eUNKNOWN_NWK`

"Standalone" (`eSTAND_ALONE`) exists in the code and is checked in many places, but **nothing ever sets it**. In practice, standalone = not joined. This document uses "not joined" for both.

1.2 Joined vs. joined and grouped

A light is **grouped** when it has at least one **multicast binding** on the light endpoint (`device_is_in_a_group()`). How it gets one:

- **Adaptiv:** auto-commissioning by the Twitch commissioning tool, or manual add (section 8.3). The light adds the group and a binding to it.
- **Daintree:** the Daintree gateway creates the group and binding.

The difference matters a lot:

	Joined, not grouped	Joined and grouped
Network status LED	red: 2 blinks every 6 s	red off
Adaptiv occupancy control	**none** (sensor runs, LED shows motion)	group occupancy (section 5.3)
Adaptiv button settings	rejected, except the always-on ones	allowed, sent to the whole group
Adaptiv daylight harvesting	no	yes (if calibrated)
Daintree occupancy	light controls itself, 2 h timeout	sensor only reports, gateway controls the light

	Joined, not grouped	Joined and grouped
Messages to other lights	none (nothing to send to)	settings and occupancy multicast to the group

1.3 Joining

- A light that is not joined keeps trying to join (network steering), with a random 0.3–8 s delay after the network goes down.
- When a light **joins from not joined**, all light settings are reset to defaults (`set_light_configurations_to_default`): levels, occupancy delays, colour, allow off, flags (section 9).
- A Daintree light can also rejoin a known network from saved commissioning data (the `B_SWITCH_NWK` path).

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2. Summary matrix

Feature	Not joined	Adaptiv joined	Adaptiv grouped	Daintree joined	Daintree grouped	Unknown
On/off and level commands	–	yes	yes	yes	yes	yes
Level =	output (≤ max)	output (≤ max)	output (≤ max)	**mapped onto min..max**	**mapped onto min..max**	output (≤ max)
Occupancy turns light on/off	yes, local	**no**	yes, group	yes, local (2 h)	**no** (reports only)	**no**
Dim to mid on vacancy	yes	–	yes	no (on/off only)	–	–
Occupancy LED (green)	yes	yes	yes	yes	yes	yes
Daylight harvesting	no	no	yes	no	no	no
Button settings (D×n, T×n)	yes	rejected	yes (group)	rejected	rejected	rejected
Colour hold (T)	yes	no change	yes (group)	no change	yes (group)	no change
Manual dim hold (D)	yes	yes	yes	yes	yes	yes
50 % / EM / factory reset / version	yes	yes	yes	yes	yes	yes
Manual add (T×6+D×6, T+D hold)	rejected	**yes**	rejected	rejected	rejected	rejected
Allow off	–	–	–	yes	yes	–
No-comms fallback (30 min → full)	–	–	–	yes	yes	–

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3. Power up

1. Both LEDs on while the firmware initialises, then off.
2. The PWM starts at the saved max level and saved colour.
3. The light goes to its **max level** 1 s after start-up (`save_max_light_level` at init schedules the move): the trim max on Adaptiv and not joined, Zigbee level 255 (= the Daintree max) on Daintree (see 4.2).

4. Occupancy starts in the ON state with a full timer. **No multicast** is sent at power up or on joining.
5. Light sensor: readings start; daylight harvesting control is suspended for the first 20 s.
6. Network status LED: first check 750 ms after init.
7. Adaptiv auto-commissioning: after a **power cycle** (not a soft reset), 1–3 s later, a light that was told to join a group by the Twitch adds that group (section 8.3).

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4. Light level control

4.1 Commands

The light accepts standard Zigbee **On/Off** (on, off, toggle) and **Level Control** (move to level, move, step, stop, with and without on/off) on endpoint 1, on every network type.

- **Adaptiv and unknown networks:** the level is the light output. Commands are limited to **25 .. max** (25 = `MIN_LEVEL`, 10 %). Dimming down stops at 25 and stays on; only an off command turns the light off.
- **Daintree:** see 4.2.

4.2 Daintree level mapping

Daintree writes three attributes in its configuration cluster (0xFC01, manufacturer 0x103A): **min** (0x0010), **max** (0x0011) and **allow off** (0xFC01).

- Writes are **stored exactly as written** (attribute, flash and internal value) and **never move the light**. The new range takes effect on the next on/off or level command.
- The Zigbee level **1..255 maps in a straight line onto min..max**: `output = min + (level − 1) × (max − min) / 254`, rounded. Example with min 20, max 200: level 255 → 200, 254 → 199, 128 → 110, 1 → 20.
- Level 255 is stored as 255 in CurrentLevel (and OnLevel). With min 0 and max 255, every level drives the light at that level.
- **Level 0 = off.** Move to Level *with On/Off* to 0 turns the light off. **Level 1 stays on** (at min). Move and Step down stop at 1.
- A plain Move to Level (without on/off) to 0 makes the light dark but leaves the OnOff attribute on (ZCL behavior).
- CurrentLevel, OnLevel and the min/max attributes always hold the Zigbee values; the mapping only affects the PWM output.
- **Allow off = false:** off commands are blocked; the light goes to level 1 (= min) and stays on.
- **No communication for 30 minutes:** the light goes to full (level 255 = max) over 1 s, once, until communication resumes.

4.3 PWM output

- Two LED channels (warm and cool). The output level is split between them by colour temperature: 3000 K = all warm, 4000 K = 50/50, 5000 K = all cool (500 K steps).
- The light driver is unstable below PWM 10, so **every channel that is on gets at least 10**; 0 stays off. A channel with a share of the mix never rounds down to 0 while the light is on.

4.4 Light settings

Setting	Default	Set by
Max (trim)	204 (80 %)	T×1–6, Twitch, Daintree (0x0011)

Setting	Default	Set by
Mid (dim level)	51 (20 %)	T×1–6, Twitch
Min	0	T×1–6, Twitch, Daintree (0x0010)
Colour	4000 K	T hold, Twitch
Allow off	true	Daintree (0xFC01)

In an Adaptiv group, settings changed with the buttons are multicast to the whole group.

4.5 Modes

- **50 % mode (T×2 + D×2):** halves max, mid and min (not below 25 for max and mid) and halves the present level. Toggling again restores the saved levels. Works on every network type.
- **EM mode (T×4 + D×4):** after vacancy the light stays at **mid instead of turning off**. Occupancy still wakes it. Works on every network type.

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5. Occupancy sensor

5.1 Sensor and LED (all networks)

- PIR sensor sampled every **75 ms**. Motion = the change from the running average exceeds the **threshold** (default 80; D×11 = 80, D×12 = 60, lower is more sensitive). A reading outside the valid range marks the sensor as bad.
- **Green LED:** on while motion is detected, off 1.25 s after the last detection. A bad sensor makes green flicker. Green blink patterns (firmware version) take priority; occupancy resumes after them.
- **Light sensor readings and green:** each lux reading needs green off for about 181 ms. While green is lit, a real reading is taken only once per 20 s (one short blink every 20 s while occupied); otherwise the last reading is reused.
- The sensor's occupancy attribute is reported on endpoint 2.

5.2 When occupancy does nothing

Condition	Effect
Unoccupied delay = 0 (D×9, or a delay write)	Sensor **stopped**. Green off. The light stays where it is; nothing turns it on or off. A stopped sensor reports "occupied".
Daylight harvesting calibration, firmware version display (`B_DISABLE_OCC_SENSOR`)	Sensor still samples and green follows motion, but occupancy does not control the light, and Adaptiv group messages are ignored.
Adaptiv, not grouped	No occupancy control (green still follows motion).
Daintree, grouped	Sensor only reports; it never turns the light on or off.
Unknown network	No occupancy control.
Short pauses after every level change (Adaptiv)	Occupancy ignored for the transition time + 1 s (2–3 s after on/off commands), so the light's own change is not seen as motion.

5.3 Adaptiv, grouped: group occupancy

Every light keeps its own timer; the group stays in sync with two multicasts (Adaptiv manufacturer-specific, occupancy sensing cluster 0x0406):

- **Keepalive (0xDF):** time until expiration (s, 2 bytes), state, reason (wake / refresh), level.
- **Level change (0xE0):** state (ON / MID / OFF), level.

A light ignores its own multicasts, and ignores these messages when its sensor is off (delay 0) or during calibration.

States: ON → *(dim delay, no motion)* → MID → *(off delay, no motion)* → OFF.

Event	What happens
Motion while ON	Restart own timer. Nothing sent.
Motion while MID or OFF	Light goes ON at once (to the wake level), and a **wake** keepalive goes out within 0–300 ms (cancelled if another light's wake arrives first).
Near the end of the ON timer (last 60 s, or timeout/4 if shorter)	One light sends a **refresh** keepalive with its remaining time, at a random point in the first 2/3 of that window. Others cancel theirs and sync their timers to it.
A light has more than the margin (window/4, 15 s) more time left than a received refresh	It replies within 0–5 s with its own (longer) time: someone near it is still there. The group syncs to the longer time and stays on.
ON timer expires	Random 0–5 s wait, then the light dims to mid and sends **level change MID**, unless another light's command arrives first (then it follows that). A light that missed the command changes on its own timer anyway.
MID timer expires	Same, with **level change OFF**: off, or stays at mid in EM mode or if min > 0. No keepalive is sent before off.
Keepalive received while ON	Timer sync only. **The level is not changed** (daylight harvesting keeps its level).
Keepalive received while MID or OFF	Light goes ON (someone has occupancy).
Level change received while seeing motion right now	Light stays ON and sends a wake, bringing the group back.
Switch on/off or level command	The ON timer restarts (all lights get the command, so they stay in sync). No message.

Wake level: the last level set from a switch (a network level command that finished or was stopped), trim max if none. After 3 hours off, the wake level returns to max.

Message count (simulation): a vacant group sends about one refresh, one dim and one off multicast per vacancy cycle. While occupied, about two multicasts per dim period (one refresh and one reply).

Special cases (Adaptiv):

Case	Behavior
Off from a switch	Occupancy may not turn the lights back on for **5 minutes**; every motion restarts the 5 minutes.
Manual-on mode (`B_MANUAL_ON_MODE`)	Occupancy never turns lights on from OFF (a switch must). It still wakes them from MID.
EM mode, or min level > 0	Vacancy ends at mid instead of off.
2-hour delay (D×10)	Temporary: returns to the saved delay on the next on/off command or once the lights are off.
Settle time after off	Occupancy can turn lights on 250 ms after they turned off.

5.4 Not joined: local occupancy

The light controls only itself, with the same delays as Adaptiv (dim and off delays from the D button), and no messages.

State	Next
ON (max level)	No motion change for 3 s and no motion → waiting
Waiting	Motion → back to **max**. No motion for the **dim delay** → dim to **mid**
Mid	Motion → **max**. No motion for the **off delay** → **off** (or stays at mid in EM mode or if min > 0)
Off	Motion → **max** (after 250 ms)

It always returns to **max**, not to a previous level, and there are no pauses after level changes.

5.5 Daintree: occupancy

Daintree, not grouped	Daintree, grouped
The light controls itself: motion → on (OnLevel); no motion for **2 hours** → off. No mid step.	The sensor **only reports** occupancy; the gateway controls the light. The occupancy attribute changes after the **PIR occupied→unoccupied delay** (endpoint 2 attribute, set by Daintree; 60 s if it can't be read).

The D-button delays do not apply on Daintree (button settings are rejected).

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6. Daylight harvesting (Adaptiv only)

- Runs only when **all** of these are true: Adaptiv network, grouped, calibrated with a lux reference, light on, occupancy state ON (not mid or off), not suspended, level above the minimum allowed, and this light is the group's **master** light.
- The master averages its lux readings (8 readings over 20 s), compares the mean with the calibrated reference, ignores errors under 5 %, and otherwise ramps the group's level over 18.5 s. The other lights follow and share their readings.
- Calibration is started from the Twitch; occupancy is suspended while it runs.
- Not used on Daintree, unknown or not-joined lights (readings are still taken; on Daintree the lux value is reported as an attribute).

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7. LEDs

Indication	LED	Pattern
Starting up	both	on during init, then off
Not joined, first check after boot	red	one long blink (2 s)
Not joined	red	1 blink (500/500 ms) every 6 s
Joined, not grouped	red	2 blinks every 6 s
Joined and grouped	red	off (checked every 12 s)
Button setting accepted	red	1 blink per press, 250/250 ms
Button input rejected	red	1 long blink (1.5 s)
Manual dim step	red	1 blink (500/50 ms) per step
Press sequence finished	both	off

Indication	LED	Pattern
Factory reset running	red	solid until the reset is done
Firmware version	red, green, red	major, middle, minor digit, 500/500 ms
Motion	green	on, off 1.25 s after the last motion
Bad occupancy sensor	green	fast flicker

Priorities: factory reset > button feedback > network status (button feedback interrupts a status blink after a 500 ms gap). Green blink patterns > lux reading > occupancy.

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8. Buttons

Two buttons: **T** (button 0) and **D** (button 1, nearest the sensor).

- **Press sequences:** press and release; the sequence ends **1.5 s** after the last release. Presses are debounced (75 ms). Both buttons can be part of one sequence.
- **Holds:** start after **4 s** of holding.
- **Feedback:** accepted = one red blink per press; rejected = one long red blink.

8.1 Press sequences

Always available (any network):

Presses	Action
T×10 + D×10 (±1 each)	**Factory reset** (section 10)
T×2 + D×2	Toggle **50 % mode**
T×4 + D×4	Toggle **EM mode**
T×8 + D×8	Show **firmware version** on the LEDs
T×6 + D×6	**Local auto-commission** (manual add + announce): **Adaptiv, not grouped only**; rejected elsewhere

Only when settings are allowed: not joined, or Adaptiv grouped. Rejected on Adaptiv not grouped, Daintree and unknown networks. In an Adaptiv group the new setting is multicast to every light in the group.

D presses (T = 0)	Dim delay (ON → mid)	Off delay (mid → off)
1	10 min	5 min
2	20 min	5 min
3	30 min (default)	5 min
4	45 min	5 min
5	60 min	5 min
8	30 s (test)	30 s
9	0 = **occupancy sensor off**	31 s
10	2 h (temporary, see 5.3)	5 min
11	sensitivity: threshold 80 (normal)	–
12	sensitivity: threshold 60 (more sensitive)	–

D presses (T = 0)	Dim delay (ON → mid)	Off delay (mid → off)
6, 7	rejected	

T presses (D = 0)	Max	Mid	Min
1	100 % (255)	20 %	0
2	80 %	20 %	0
3	60 %	20 %	0
4	40 %	20 %	0
5	20 %	10 %	0
6	10 %	10 %	0

Any other combination is rejected.

8.2 Holds

Hold	Action	Where
T held 4 s	Colour temperature +500 K every 2.5 s (3000 → 5000 K, then wraps). Multicast to the group.	Changes colour only when grouped (Adaptiv or Daintree) or not joined; elsewhere the hold runs but nothing changes.
D held 4 s	Manual dim: steps down about every 200 ms; below 25 it jumps back to max.	Every network type
T + D held 4 s	Manual add to group (8.3), repeats every 4 s while held	**Adaptiv, not grouped only**

8.3 Adaptiv commissioning (auto and manual add)

- **Auto-commissioning (Twitch):** the Twitch sends an "announce group" command with a group ID. The light saves it, and after the **next power cycle** (1–3 s after start-up) adds that group.
- **Manual add mode** (T+D hold, or a Twitch command): the light ramps up and down continuously until it is added to a group, or the time-out expires (5 min by default, or the Twitch's value). Being added to a group stops it and returns the light to max.
- **Local auto-commission** (T×6 + D×6): manual add mode, plus the light broadcasts the announce itself 3 times, 3 s apart (doing the Twitch's job).

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9. Defaults

Applied on joining from not joined, and by factory reset.

Setting	Default
Max / mid / min level	204 (80 %) / 51 (20 %) / 0
Dim delay (ON → mid)	30 min
Off delay (mid → off)	5 min
Unoccupied → occupied delay	0 s (250 ms settle)
Occupancy threshold	80

Setting	Default
Colour	4000 K
Allow off	true
Lux target	0 (daylight harvesting off until calibrated)

10. Factory reset

T×10 + D×10 (±1). Red LED solid until done. Steps, 500 ms apart:

1. Leave the network.
2. Clear the reporting table, groups, bindings and scenes.
3. Reset settings to defaults (section 9) and the network type to not joined. A normal reset also clears the mode flags (50 %, EM, manual-on, calibration, 2 h delay, manual add).
4. Restart the chip.

11. Observations and open questions

Behaviors in today's code that look unintended or need a decision:

1. ** `eSTAND_ALONE` is never set.** Standalone and not joined are the same today; the separate standalone branches in the code are unused.
2. **Unknown networks have no occupancy control.** Occupancy is initialised as for Daintree, but the sample handler has no case for unknown networks, so the sensor never turns the light on or off.
3. **Daintree max defaults to 204 (80 %)** after joining, until the gateway writes its own max. Until then, level 255 maps to 80 %.
4. **D×9 sets an off delay of 31 s** with the sensor off; the 31 has no effect.
5. **50 % mode, EM mode and manual dim work on Daintree**, while the other button settings are rejected there. On Daintree, 50 % mode halves the Daintree min/max range.
6. **Daintree grouped lights never control themselves from occupancy;** the gateway must. Not-grouped Daintree lights use a fixed 2-hour timeout.
7. **Not-joined occupancy always wakes to max**, while Adaptiv wakes to the last switch level.
8. **A stopped sensor (delay 0) reports "occupied"** in its attribute.
9. **The colour hold on a not-grouped Adaptiv or Daintree light** runs (no feedback) but changes nothing. On a grouped Daintree light it changes the colour and multicasts it to the group.